

Analysis of the enterprise financial performance evaluation based on deep convolutional neural network

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Abstract: When evaluating the financial performance of an industry or a company, since each industry has its own unique industry characteristics, only by selecting appropriate indicators can a sound and effective evaluation mechanism be established, and then the industry or enterprise can be evaluated. Carry out more in-depth positioning and awareness. This paper selects the 5-year financial data of 144 listed companies, conducts preliminary selection of indicators on the basis of sorting out relevant research at home and abroad, and constructs a financial evaluation indicator system for listed companies. The premise of establishing a listed financial evaluation model is to classify listed companies. To this end, this paper analyzes a variety of performance measurement methods. The CRITIC method and the entropy method are selected to weight the three performance measurement methods of the TOPSIS method, the grey relational analysis and the fuzzy comprehensive evaluation, to calculate the financial performance score of the company, and to cluster its scores. In the means clustering, SOM clustering is preferred to label the input data of the convolutional neural network. Then build a convolutional neural network model for evaluation and simulation training. It is verified whether the convolutional neural network model constructed in this paper can evaluate the performance of Chinese listed companies well, and provide some valuable references and warnings for the development of Chinese listed companies. Finally, the convolutional neural network model and the support vector machine model are compared and analyzed. The research results show that the GRA-Entropy-SOM-CNN model constructed in this paper has a good evaluation effect, and the highest accuracy rate can reach 97.18%. Evaluation of financial status of listed companies.

Key words: convolutional neural network; listed company; financial performance evaluation

1 INTRODUCTION

Under the environment of rapid economic growth in my country, enterprises have also achieved rapid and long-term development. The development of Internet technology and the improvement of domestic innovation capabilities have enabled enterprises to

obtain a new development environment and growth model. The industry has a wide audience, a huge number of customer groups, and the demand and consumption of products are considerable. In addition, with the gradual integration of traditional enterprises and Internet technology, more diversified business models have emerged, which has promoted the transformation and development of enterprises. While ushering in opportunities, they also face challenges ^[1].

The evaluation of the company's financial performance can help the company to discover its own problems. The evaluation results represent the operating results of the company to a large extent, and the level of the score means the quality of the operation. At the same time, this intuitive reflection is also conducive to the comparative analysis of enterprises and the same industry ^[2]. The managers of the enterprise improve the management system of the enterprise through financial performance evaluation, and promote the growth and upgrading of the enterprise. A good level of financial performance represents that the enterprise has a good management model and operating performance, which reduces the performance of safety problems and promotes the benign and sustainable development of the enterprise. Therefore, the shareholders and management of the company are the main body of evaluation to evaluate the financial performance of the company. The problems existing in the process, aiming at the deficiencies, will be focused on and improved in the future decision-making. The industry belongs to the trust product industry. Consumers and investors will have insufficient confidence in the trust product industry. More intuitive judgments such as financial performance evaluation are needed to help consumers and investors judge the pros and cons of enterprises. Managers of enterprises evaluate financial performance and disclose the evaluation results in a timely manner, which is conducive to gaining the trust of consumers and helping enterprises establish a good image. In addition, evaluating the financial performance of enterprises can also provide a basis for government departments to issue relevant policies, which is conducive to the overall control of the industry by government departments ^[3].

According to the country's strategic deployment, in the new stage, the development of the industry is an inevitable requirement to meet people's needs for a better life. Relevant enterprises are placed great hopes and shoulder more responsibilities at the

same time^[4]. It is imperative to promote the transformation of enterprises and promote the development of the industry. However, the current research on the financial performance of enterprises is not enough. In order to adapt to the development of the trend, a scientific and effective enterprise financial performance evaluation model is established, and the model is used to evaluate the financial performance of the enterprise, and timely and effectively adjust the operation according to the evaluation results. The strategy has practical significance for the long-term and healthy development of enterprises^[5].

2 STATE OF THE ART

Scholars are committed to choosing a more scientific method of assigning weights to make the evaluation results more objective. Experts and scholars have gradually combined operations research, statistics, data analysis and other methods with the financial performance evaluation process, in order to make the evaluation results more scientific and reliable, and their research results are quite rich.

The AHP was produced after 1970, and its core element is to divide a complex evaluation item into orderly levels with reference to qualitative methods, and then assign them according to the relative size of each level's effect on the target according to quantitative methods. Compared with purely subjective weighting, the obtained evaluation results are more objective, and it is a comprehensive evaluation method. Hayat M. Awan and M.Ishaq Bhatti (2015) took 200 manufacturing companies as the research object, and used the analytic hierarchy process to study the financial performance management related to improving the profitability of enterprises. After comparing a variety of evaluation methods, Gu Yu and Wang Hehuan (2015) chose AHP to establish a financial performance evaluation model for the logistics industry, compared and analyzed the scores of different types of logistics companies, and analyzed the overall status of the logistics industry? 1 . AHP has certain limitations, so some scholars have improved it when applying it to financial performance evaluation. Cheng Bo and Jia Guozhu (2015) proposed that the impact of each expert's score on the final result is not equal, and it should be determined according to the actual interests of the experts. The comprehensive score is calculated after the weighted average. The improved method can Effectively avoid the problem of unfair scoring by experts due to interest-related issues^[6].

The appearance of the data envelopment method can be traced back to the 1950s, and it was formally proposed and established a model by Professors Chanes and Copper in the United States in 1978. It is a multidisciplinary field. Its main body must be comparable, and it is a relative effectiveness evaluation model. When using the data envelopment method to study the financial performance of enterprises, it is often combined with other analysis methods such as factor analysis and analytic hierarchy process. Moon Hana et al. (2020) believe that the data envelopment method is a method that can use both quantitative indicators and qualitative indicators, and DEA can complete traditional indicator analysis. By analyzing the data of 5 commercial banks for 10 years, the final research results show that the fuzzy DEA method can obtain different financial performance scores according to the degree of uncertainty when the financial index value is non-deterministic value compared with the traditional DEA method.

Regarding the research on factor analysis, Gao Yanfang and Wang Lu (2016) used factor analysis to compare and analyze the financial performance of the three giant companies in Chinese telecommunications industry. The research process found that each company's ability was weak, and the final evaluation The results show that mobile companies are the strongest of the three. Tong Yuanke and Wei Yunjie (2021) used factor analysis to extract six common factors from 11 financial indicators to calculate the financial performance score of the research object^[7].

In addition to the above methods, models such as grey relational degree method, fuzzy evaluation method, TOPSIS, and direct value method are also widely used in performance evaluation. Deng Yuwen (2012) analyzed the applicability of this method by explaining the principle of grey relational degree. The research results show that this method is suitable for both the performance evaluation of a single enterprise and the comparative analysis of enterprises in the whole industry. In order to avoid the shortcomings of the existing evaluation methods, Jin Yingwei and Zhang Min (2019) took listed publishing companies as the research object, and chose the relative priority method in the fuzzy evaluation method to evaluate financial performance. Yuan Shengjun et al. (2020) used the direct rights method to evaluate the financial performance of 32 companies with strong brand value, and analyzed the internal relationship between brand value and financial performance. Cocis Andreas Daniel et al. (2021) for the study of

airlines The relationship between reputation and financial performance, using the TOPSIS method to evaluate the financial performance of 22 airlines³⁴¹[8].

In order to establish a comprehensive and effective model, many scholars will also combine the two methods to make the empowerment process more scientific. Zhang Jing et al. (2013) combined the analytic hierarchy process and the direct value method to calculate and analyze the financial performance of energy companies from the perspectives of economy, society and environment, overcoming the lack of subjectivity of weights. Shaverdi M et al. (2014) believed that when faced with many standards and indicators, it can be regarded as a fuzzy multi-criteria decision-making problem, and linked the fuzzy evaluation method and the AHP model to the seven Iranian oil companies. Financial Performance Evaluation 36. Ksenija Mandic and Boris Delibasic (2014) conducted an analysis of the entire banking industry in Serbia with the aim of proposing a multi-criteria model, using AH to prioritize criteria and then applying the TOPSIS method to rank bank financial performance. Fan Shuhai et al. (2014) established the AHP-TOPSIS model, which uses a judgment matrix to determine the indicator weights, and then uses TOPSIS to calculate the relative proximity as the financial performance evaluation result.

The BP neural network was initially more used in prediction, and with the continuous research, many scholars have applied it to the evaluation problem. The use of BP neural network for financial performance evaluation research needs to be combined with other methods to achieve. Zhao Quanchao et al. (2005) conducted in-depth research on the structure design and function selection of BP neural network, and combined with the balanced scorecard method to construct an optimized BP neural network performance evaluation model. Xu Shengnan et al. (2011) took GEM companies as the research object and proposed the LMBP neural network performance evaluation model. Xu Nan and Xie Ruifeng et al. (2014) used correlation analysis to screen indicators in order to study the financial performance of the logistics industry, and established a neural network model for financial performance of logistics enterprises. Cai Yanping and Sun Xia (2016) introduced the EVA value to enrich the index system, used the grey correlation method to determine the performance score, and established a neural network model that conforms to the performance evaluation of listed commercial banks. Zhang Meng and Han Yujing

(2019) in the context of the Internet promoting agricultural development, BP neural network combined with AHP to build a performance evaluation model I that conforms to the characteristics of "Internet + agriculture" enterprises. Zhou Hong (2019) used the advantages of genetic algorithm to make up for the shortcomings of BP neural network. The combination of the two played a complementary effect, making the GA-BP neural network model for financial performance evaluation of listed companies more perfect.

3. METHODOLOGY

3.1 Data preprocessing

Among the 36 financial indicator systems of listed companies, there are positive indicators, neutral indicators and negative indicators^[9]. In this financial indicator system, the positive indicators are (X1, X2, X3, X4, X5, X6, X7, X8, X9, X10, X13, X14, X15, X16, X17, X21, X22, X23, X24, X25, X26, X27, X30, X31, X33, X34, X35, X36), neutral indicators are (X4, X29), negative indicators are (X11, X12, X18, X19, X20, X28, X32). Positive indicators have a positive correlation with the financial performance evaluation of listed companies, and are treated according to the positive indicators. Negative indicators have a negative correlation with them and are treated as negative indicators. Since the correlation between neutral indicators and comprehensive evaluation results cannot be measured, and the optimal value is difficult to determine, considering that among the 36 indicators, there are only 2 neutral indicators, which have little impact on the comprehensive performance evaluation results of listed companies, and the difference between the indicators The difference is also very small, so it is uniformly processed according to the positive index processing method.

Among the financial performance indicators of the 36 indicators of listed companies such as agriculture, forestry, animal husbandry and fishery, due to the different unit standards of different indicators, there are large differences in magnitude^[10]. Among them, the maximum value is 38126549.56 of the interest payment multiple (%); the minimum value is cash and cash equivalents^[11]. The net increase (10,000 yuan) is -1320310, the difference between the indicators is too large, so it is necessary to carry out dimensionless treatment of the indicators before the comprehensive evaluation performance measurement. In order to make the index system coefficients of listed

companies distributed between 0 and 1, the following dimensionless processing method is adopted:

(1) Positive indicators:

$$X_{ij} = \frac{e_{ij} - e_{j\min}}{e_{j\max} - e_{j\min}} \quad (1)$$

(2) Negative indicators:

$$X_{ij} = 1 - \frac{e_{ij} - e_{j\min}}{e_{j\max} - e_{j\min}} \quad (2)$$

The CRITIC weighting method was proposed by Diakoulaki, which determines the objective weight of each indicator through the contrast between the measurement indicators and the conflict between the evaluation indicators. The greater the standard deviation of each scheme within the same indicator, the greater the value gap between them. If there is a strong positive correlation between two financial data indicators, then the conflict between the two financial data indicators is low.

$$C_j = \delta_j \sum_{i=1}^m (1 - r_{ij}) \quad (3)$$

$$W_j = \frac{C_j}{\sum_j C_j} \quad (4)$$

Calculated by the above formula through MATLAB 2018a simulation (see Appendix A for the code), the weights of 36 indicators can be obtained as shown in Table1 below. The indicators with larger weight obtained by the CRITIC method are: total liabilities (X11), current liabilities (X12), current asset turnover rate (X31), main business profit rate (X23), and total assets (X9)^[12].

Table1 CRITIC method indicator weights

Index number	The weight	Index number	The weight
X1	0.014735	X19	0.018503
X2	0.045801	X20	0.019917
X3	0.028174	X21	0.017757
X4	0.014486	X22	0.034881

X5	0.02136	X23	0.048844
X6	0.018548	X24	0.016083
X7	0.025782	X25	0.013548
X8	0.017661	X26	0.014703
X9	0.048003	X27	0.014655
X10	0.038033	X28	0.031599
X11	0.063676	X29	0.012252
X12	0.055213	X30	0.014672
X13	0.035934	X31	0.054766
X14	0.029211	X32	0.030191
X15	0.030833	X33	0.018295
X16	0.035007	X34	0.035178
X17	0.015105	X35	0.020241
X18	0.026068	X36	0.020285

The entropy method has been widely used in various fields of social economy. It is based on the variability among indicators to determine weights^[13]. The smaller the information entropy E_j of an indicator, the greater its weight. Because the greater the degree of variation, the more information it contains, and it can play a greater role in the comprehensive performance measurement^[14].

The entropy method is used to calculate the weight, and the entropy value H_j of the n -item indicators is calculated from the formula (5), and the entropy value H_j is substituted into the formula (6) to calculate the weight of the n -item indicators:

$$H_j = -k \sum_{i=1}^m f_{ij} \ln f_{ij}, \quad (1 \leq j \leq n) \quad (5)$$

$$W_j = \frac{1 - H_j}{n - \sum_{j=1}^n H_j}, \quad (1 \leq j \leq n) \quad (6)$$

The above formula can be calculated by MATLAB 2018a simulation (see Appendix A for the code), and the weights of 36 indicators can be obtained^[15]. The indexes with larger weight obtained by the entropy method are: main business income (X2), cash ratio (X16), total assets (X9), quick ratio (X15), and current assets (X10).

3.2 Overview of Convolutional Neural Networks

The input of the convolution operation is a two-dimensional array data (identified as I , the coordinates are (m, n)), the convolution kernel corresponds to a two-dimensional data (identified as K), and the obtained feature map is also a Two-dimensional data data (identified as S , coordinates at (I, j))^[16]. In image recognition, I is a two-dimensional matrix composed of pixel values of the image, and K is a weight parameter optimized by the learning algorithm, which is also a two-dimensional matrix and can be multiple. The formula for the convolution operation is:

$$S(i, j) = (I * K)(i, j) = \sum_m \sum_n I(m, n) K(i - m, j - n) \quad (7)$$

Convolution operations have 3 important properties: sparse connections, parameter sharing, and equivariant representation^[17]. The convolution operation achieves the property of sparse connectivity by restricting the size of the kernel to be smaller or much smaller than the size of the input. This is different from the full connection method. Through sparse connection, the purpose of reducing the number of weight parameters can be achieved. On the one hand, the computational complexity is reduced, and on the other hand, the over-fitting phenomenon caused by too many connections is reduced, and the model is improved. generalizability. Parameter sharing is when the same parameter is used in multiple functions of the same model. Parameter sharing in the convolution operation also effectively reduces the number of parameters. Equivariance in a mathematical function has the characteristic that when the input changes, the output changes in the same way. For example, for a function $f(x)$, if there is a function $g(x)$ such that $f(x)$ satisfies $f(g(x))=g(f(x))$, we say that $f(x)$ is for variable g Equivariant. The translation equivariance of convolution is a useful property, when processing images, if you move some objects in the input, the features in the output will also move a certain amount. Parameter sharing is a prerequisite for realizing translation equivariance^[18].

In most cases, we want to extract multiple features, which needs to be achieved by increasing the number of convolution kernels. As long as there are enough convolution kernels, enough features can be extracted^[19]. The length, width and depth of the convolution kernel required by the convolution layer need to be manually specified. The length and width of the more commonly used filters are $3*3$ and $5*5$ ^[20].

After the features are obtained through the convolution operation, these features need to be used for classification. Generally speaking, the features extracted by the convolution operation can be used as the input of the classifier, but doing so will face a huge amount of calculation, so the pooling operation is needed to further process the feature map results obtained by the convolution operation. The pooling function used in the pooling operation can statistically summarize the eigenvalues of a certain position in the plane and its adjacent positions, and use the summary result as the value of this position in the plane. For example, the max pooling function will calculate the maximum value within the position and its adjacent rectangular area, and use this maximum value as the value of the position. The size of the feature map after the pooling operation is further reduced. The process of pooling operation is similar to that of convolution operation.

3.3 Establishment of Convolutional Neural Network Model

Figure 1 shows a relatively simple and specific convolutional neural network architecture diagram for solving classification problems. It can be seen from the figure that a convolutional neural network mainly includes five structures: input layer, convolutional layer, pooling layer, fully connected layer and output layer.

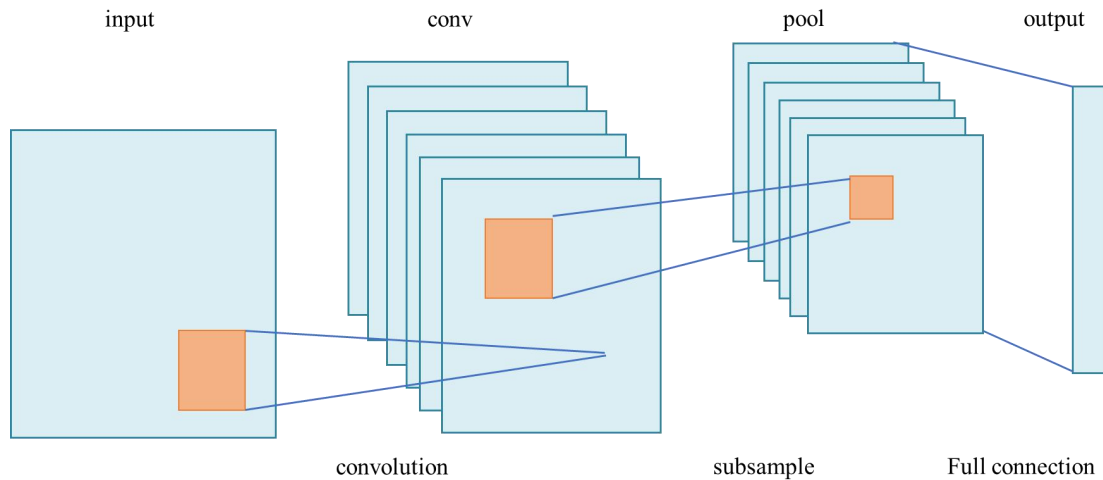


Figure 1 Convolutional neural network structure

Substitute 710 pieces of data into the convolutional neural network model for simulation training. The input layer is 36 vectors with dimension 6×6 . In the convolutional neural network, the input layer is 36 index data of any company, which are input neurons, and then processed by neurons in several hidden layers, and finally the

classification information of listed companies is directly output in the output layer. During this period, the entire neural network undergoes a lot of weight adjustment.

The convolution kernel size of the convolutional layer in this paper is 5*5, the depth is 6, and the stride is 1 without all 0 padding. In the convolutional layer, the function f is the input function, the function g is the convolution kernel, and the asterisk * represents the convolution. The superposition of these two functions results in a feature map. The convolution operation can extract features from the input block matrix and preserve the relative relationship between them. A local area (that is, a small matrix) is used to scan the entire data matrix. Under the action of this local small matrix, all data points of the large matrix will be linearly transformed and combined to form the neuron nodes of the next layer. This local area is is the convolution kernel.

$$s(i, j)=f(i, j) * g(i, j) \quad (8)$$

Try to use 1*1, 3*3, 5*5 for the size of the convolution kernel, and the depth of the convolution kernel is 1~7 layers. The simulation test results are shown in Table2. When the size of the convolution kernel is 5*5, the model's The accuracy is generally better than 1*1 and 3*3. Among them, when the convolution kernel size is 5*5 and the depth is 6, the accuracy rate is the best, which is 90.91%.

Table2 Comparison of the size and number of different convolution kernels

Convolution kernel size	1*1	1*1	1*1	1*1	1*1	1*1	1*1
Convolution kernel number	1	2	3	4	5	6	7
accuracy	61.36%	70.45%	79.55%	61.36%	68.18%	75%	75.00%
Convolution kernel size	3*3	3*3	3*3	3*3	3*3	3*3	3*3
Convolution kernel number	1	2	3	4	5	6	7
accuracy	72.73%	70.45%	77.27%	72.73%	77.27%	88.64%	84.09%
Convolution kernel size	5*5	5*5	5*5	5*5	5*5	5*5	5*5

Convolution kernel number	1	2	3	4	5	6	7
accuracy	75.00%	84.09%	84.09%	81.82%	84.09%	90.91%	79.55%

In this layer, the matrix multiplication between the input vector and the bias parameter is calculated. Each unit also adds a bias term, and finally passes it to the output layer through the Sigmoid activation function. It can limit the upper bound of the value range of the input data, and will not The neurons in the neural network cannot update the weights because the learning rate is too large.

$$\text{sigmoid}(x) = \frac{1}{1 + e^{-x}} \quad (9)$$

The OUTPUT layer is also a fully connected layer with a total of 5 units, and these 5 units correspond to the classification probabilities of financial performance levels of 1 to 5 respectively. The criterion is that if a unit outputs 0 or close to 0, then the position of the unit in this layer is the number recognized by the network. This output is produced because the unit at this layer computes the radial basis function:

$$y_i = \sum_j (x - w_{i,j})^2 \quad (10)$$

The calculation of the RBF is related to the bitmap encoding of the i th class. For the i th unit, the closer the value of y_i is to 0, the closer it is to the bitmap encoding of the i th number, that is, the recognition result of the current network input is the i th category. The hierarchical division of financial performance clustering is the data label of the output layer of the convolutional neural network.

4 RESULT ANALYSIS AND DISCUSSION

4.1 Simulation Research on Convolutional Neural Network Financial Performance Evaluation Method

The convolutional neural network model constructed in this paper uses 710 pieces of data for training, and the ratio of training set and test set is 4:1, that is, 568 pieces of data in the training set and 142 pieces of data in the test set. The convolution kernel size of the convolution kernel is 5*5 and the depth is 6. The iterative descent rate is initially 1, and can be set to gradually decrease with the number of iterations during the training process.

The batch sample size is 2. The size of the input data matrix is 41×710 , and each piece of data contains 36 items of indicator data and 5 items of label data. The training set for each training is randomly selected 568 pieces of data, and the remaining data is used as the test set.

The accuracy of TOPSIS-Entropy-SOM-CNN and TOPSIS-CRITIC-SOM-CNN methods is shown in Figure 2. It can be clearly seen that the entropy weighting method has a higher accuracy. At 1050 iterations, the TOPSIS-Entropy-SOM-CNN method achieves the highest accuracy of 95.07%. The average accuracy of the TOPSIS-Entropy-SOM-CNN method in the interval of 500-1500 iterations is 88.63%, and the average accuracy of the TOPSIS-CRITIC-SOM-CNN method in the interval of 500-1500 iterations is 84.10%. It can be seen that The TOPSIS-Entropy-SOM-CNN method is significantly better than the TOPSIS-CRITIC-SOM-CNN method.

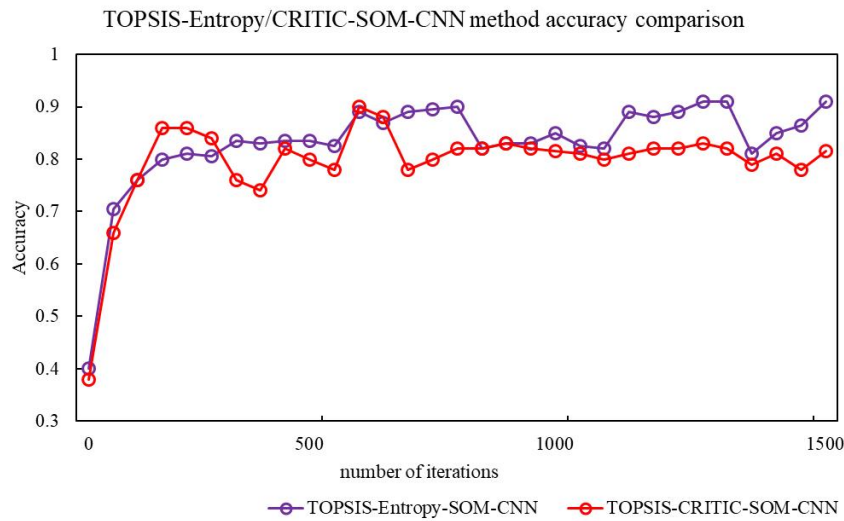


Figure 2 Comparison of the accuracy of the TOPSIS-Entropy/CRITIC-SOM-CNN method

The accuracy of GRA-Entropy-SOM-CNN and GRA-CRITIC-SOM-CNN methods is shown in Figure 3. It can be clearly seen that the entropy weighting method has higher accuracy. At 700 iterations, GRA-Entropy-SOM-CNN achieves the highest accuracy of 97.18%. The average accuracy of GRA-Entropy-SOM-CNN in the range of 500-1500 iterations is 91.72%, and the average accuracy of GRA-CRITIC-SOM-CNN in the range of 500-1500 iterations is 79.61%. It can be seen that GRA-Entropy-SOM-CNN is significantly better than GRA-CRITIC-SOM-CNN method.

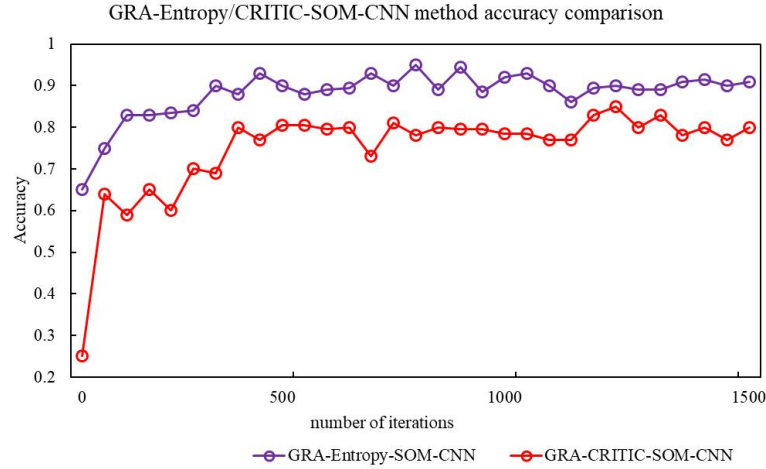


Figure 3 Comparison of accuracy of GRA-Entropy/CRITIC-SOM-CNN method

The accuracy of the FCE-Entropy-SOM-CNN method and the FCE-CRITIC-SOM-CNN method is shown in Figure 4. It can be clearly seen that the entropy weighting method has higher accuracy. When the number of iterations is 1200, the highest accuracy of the FCE-Entropy-SOM-CNN method is 91.55%. The average accuracy of FCE-Entropy-SOM-CNN in the range of 500-1500 iterations is 86.45%, and the average accuracy of FCE-CRITIC-SOM-CNN in the range of 500-1500 iterations is 82.60%. It can be seen that FCE-Entropy-SOM-CNN is significantly better than FCE-CRITIC-SOM-CNN method.

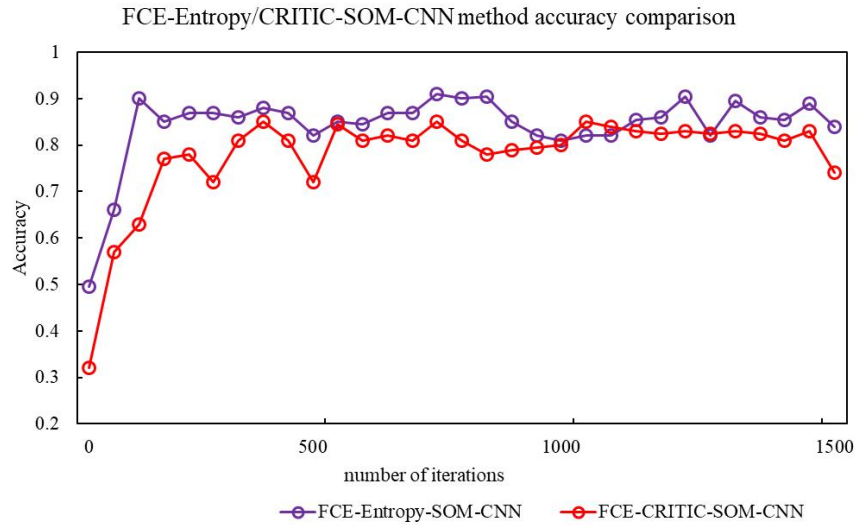


Figure 4 Accuracy comparison of FCE-Entropy/CRITIC-SOM-CNN method

4.2 Comparative analysis of financial performance evaluation models and simulation experiments

Comparing Figure 5 and Figure 6, it can be seen that the accuracy of index weighting using the entropy method after training and prediction by the convolutional neural network model is significantly higher than that of the CRITIC weighting method, indicating that the entropy method can be used as a performance measurement method for listing performance evaluation. more accurate.

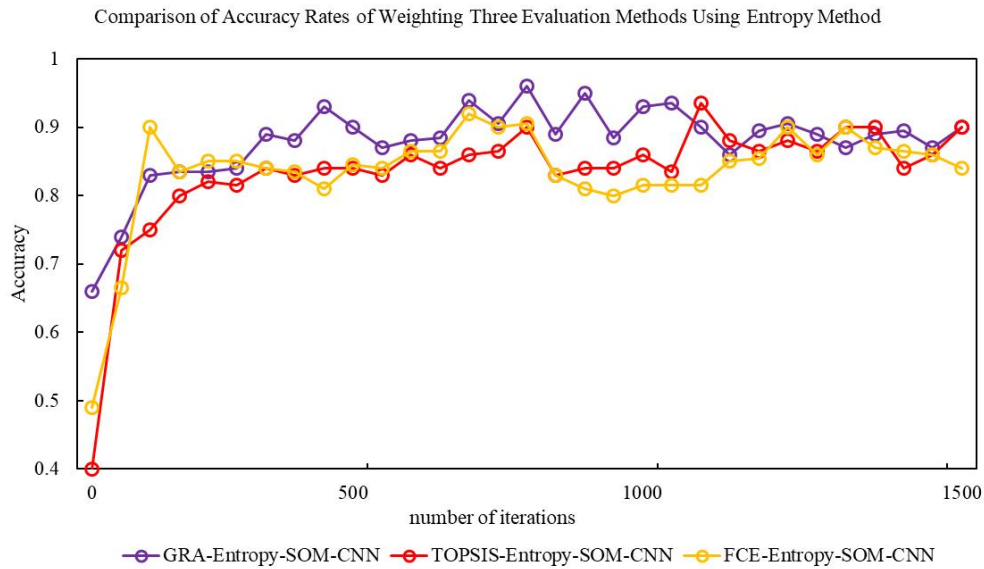


Figure 5 Weighting by entropy method

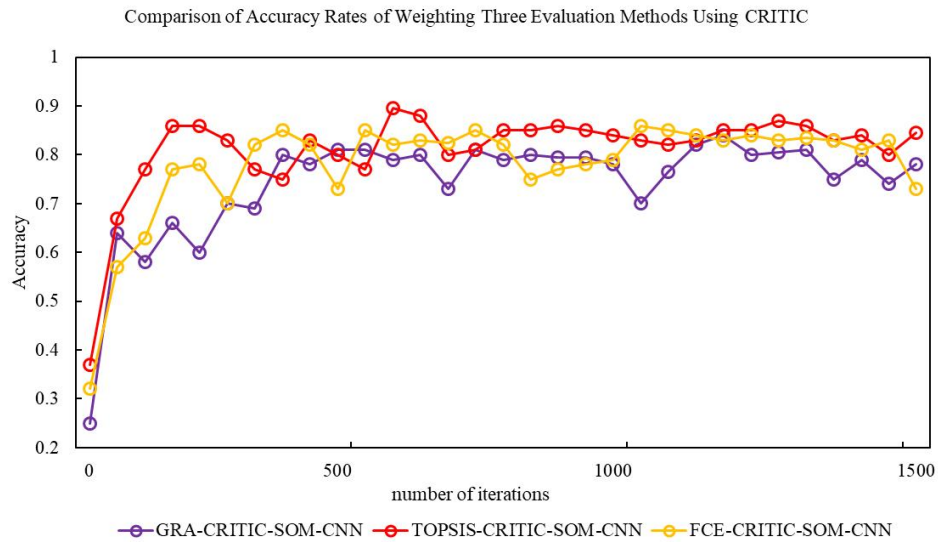


Figure 6 CRITIC Law Empowerment

As shown in Table3, the GRA-Entropy-SOM-CNN simulation predicts the best. The highest accuracy of 97.18% and the average accuracy of 91.72% are the highest among all methods. Although the accuracy of other models is not as good as GRA-Entropy-SOM-CNN, they also have better prediction effects, indicating that the listing performance evaluation model based on convolutional neural network constructed in this paper can well predict and classify the financial performance of listed companies .

Table3 Accuracy comparison

	TOPSIS-Entropy-SOM-CNN	TOPSIS-CRITIC-SOM-CNN	GRA-Entrop-SOM-CNN	GRA-CRITIC-SOM-CNN	FCE-Entrop-SOM-CNN	FCE-CRITIC-SOM-CNN
Highest accuracy	95.07%	39.44%	97.18%	85.21%	91.55%	88.03%
Average accuracy	8.63%	34.10%	91.72%	79.61%	86.45%	82.60%

In order to verify the effect of convolutional neural network in deep learning, this paper also builds an SVM model for comparison. Substitute the same data and labels as the GRA-Entropy-SOM-CNN model into the SVM model, set the Kernel scale mode to automatic, and the Multiclass method to One-vs-One. The classification accuracies of Linear SVM, Quadratic SVM and Cubic SVM are shown in Table 4 below. When the Box constraint level is 3, Linear SVM has the highest accuracy of 93%.

Table 4 SVM model accuracy comparison

Box constraint level	Linear SVM	Quadratic SVM	Cubic SVM
1	91.70%	90.08%	88.30%
2	92.70%	90.40%	89.70%
3	93.00%	90.40%	89.60%
4	92.80%	90.40%	89.60%

In the best case, the accuracy of GRA-Entropy-SOM-SVM is 93%, which is not as good as the accuracy of convolutional neural network evaluation classification, as shown in Table 5. It shows that the convolutional neural network model constructed in this paper has a better evaluation and classification effect than the support vector machine model.

Table 5 Comparison of CNN and SVM accuracy

Early warning model	Accuracy
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GRA-Entropy-SOM-CNN	97.18%
GRA-Entropy-SOM-Linear SVM	93.00%

5 CONCLUSION

The three objective weighting methods of CRITIC method, direct value method and coefficient of variation method are compared. The idea of constructing weights between the direct value method and the coefficient of variation method is relatively close, and the indicators that account for a larger weight are basically the same. The CRITIC method is different from the other two in that the weighting of the indicators is relatively average, and there is no big difference between the indicators weighted by the direct value method. Comparing the top-ranked indicators, it can be found that the two indicators of main business income and total assets have a large weight in the three methods, which have a greater impact on the performance measurement of listed companies. A combination of three financial performance measures and two weighting methods. The degree of financial performance of the six combined methods (TOPSIS-Entropy, TOPSIS-CRITIC, GRA-Entropy, GRA-CRITIC, FCE-Entropy, FCE-CRITIC) was obtained. GRA-Entropy and TOPSIS-Entropy obtained 3 out of the 5 companies with higher financial performance, which are the same. There are two different companies. The company obtained by GRA-Entropy method is ST company, and the company obtained by TOPSIS-Entropy method is non-ST company. Compared with TOPSIS-Entropy method, GRA-Entropy method The law will be more accurate. Further observing the GRA-Entropy method and the GRA-CRITIC method, comparing the top and bottom companies, it can be found that the CRITIC empowerment method cannot well highlight the impact of financial indicators on company performance. The deep learning-based convolutional neural network constructed in this paper has good results. When the convolution kernel size is 5×5 , the depth is 6, and the number of iterations is 700, the accuracy of the GRA-Entropy-SOM-CNN model in this paper can reach 97.18%. At the same time, the convolutional neural network model constructed in this paper has a better prediction effect than the SVM model. To a certain extent, it can provide financial evaluation for listed companies. At the same time, it also shows that the gray direct correlation analysis method and the SOM neural network can measure and cluster the financial performance of listed companies very well.

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